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DATA 71200: Advanced Data Analytic Methods

FINAL EXAM (Spring '26)

Instructions

This exam is comprised of six (6) statistical problems and accompanying datasets. To complete the exam, choose any four (4) problem sets and address the questions posed by each one. For each problem prepare a thorough and responsive analysis including: (a) a statement of the hypotheses you tested; (b) a description of and rationale for your choice of statistical analytic approach; (c) summarize and present the results of your statistical analyses; and (d) provide an interpretation of your findings and the conclusions you reached. Each problem is worth 25 points (for a total exam score of 100 points).

Each problem has three parts that move from conceptual to applied, reflecting your stated goal of developing statistical thinking rather than rote procedure. The point allocations weight analytic and interpretive work over computation alone.

Please indicate the question numbers of your responses and submit them in one extended *MSWord* file and include screenshots of your code. Show all your work in one inclusive file and attach it to an email to me (HEverson@gc.cuny.edu) using the subject line: DATA71200 Spring '26 Final Exam. Please submit your final exam no later than 26 May 2026.

Problem 1. Predicting Housing Prices with Multiple Regression

You are given a dataset of 500 residential property sales containing predictors: square footage, number of bedrooms, neighborhood median income, age of structure, and distance to the nearest transit hub. The outcome variable is sale price (in $000s).

- a. Describe the data preprocessing steps you would perform before fitting a regression model. Explain your reasoning for each step. (5 pts)
- b. Fit a multiple regression model predicting sale price. Report and interpret the coefficients for at least three predictors, including a discussion of statistical significance and practical significance. (10 pts)
- c. Evaluate model fit using appropriate diagnostics (e.g., R^2 and residual plots). Identify and discuss any violations of regression assumptions you observe. (10 pts)

DATA Source: The Ames Housing Dataset (De Cock, 2011) contains 79 potential explanatory variables describing residential properties in Ames, Iowa, including square footage, neighborhood, year built, and sale price. The original article is: De Cock, D. (2011). Ames, Iowa: Alternative to the Boston housing data set. *Journal of Statistics Education*, 19(3). The data are available for download at: https://www.kaggle.com/datasets/prevek18/ames-housing-dataset and/or through the openml Python library (dataset_id=42165).

Problem 2. Binary Classification of Graduate Admissions

A university admissions dataset contains GRE scores, undergraduate GPA, institution prestige rank, and a binary outcome: admitted (1) or not admitted (0). Develop and fit a model of the probability of admission.

- a. Explain why ordinary least squares regression is inappropriate for this outcome and how logistic regression addresses the problem. (5 pts)
- b. Fit a logistic regression model. Interpret the odds ratio for GRE score and prestige rank. What do these values tell an admissions officer? (10 pts)
- c. Construct and interpret a confusion matrix. Calculate accuracy, precision, and recall. At what classification threshold would you operate this model, and why? (10 pts)

DATA Source: The UCLA Graduate Admissions Dataset contains GRE score, GPA, institutional prestige rank (1–4), and a binary admit outcome for 400 applicants. These data were originally distributed by *UCLA's Institute for Digital Research and Education*. Available directly at: https://stats.idre.ucla.edu/stat/data/binary.csv and on the Kaggle site as "Graduate Admissions."

Problem 3. Bayesian Updating for Customer Conversion Rates

A marketing team believes a new product page has a 20% conversion rate. After running an A/B test, they observe 45 conversions in 150 visits. Use a Beta-Binomial model to analyze the updated evidence.

- a. Specify a reasonable prior distribution consistent with the team's prior belief. Justify your choice of hyperparameters. (5 pts)
- b. Derive and describe the posterior distribution given the observed data. How has the evidence shifted belief? (10 pts)
- c. Contrast the Bayesian credible interval for the conversion rate with a classical 95% confidence interval computed from the same data. Discuss the interpretive differences for a non-technical stakeholder. (10 pts)

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DATA Source: Use the Cookie Cats Mobile Game A/B Test Dataset on Kaggle (Huusom & Heide, 2020), which contains retention outcomes across two experimental conditions and maps naturally onto Beta-Binomial updating. Available at: https://www.kaggle.com/datasets/yufengsui/mobile-games-ab-testing

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Problem 4. Supervised Learning — Classification

Classification Algorithms on Imbalanced Data

You have a medical dataset of 10,000 patients, of whom 8% have been diagnosed with a rare condition. Your task is to classify patients using k-Nearest Neighbors, a Naïve Bayes classifier, and a Decision Tree.

- a. Describe the class imbalance problem in this dataset. What risks does it create for model evaluation, and how would you address it in practice? (5 pts)
- b. Using 5-fold cross-validation, compare the three algorithms on F1-score and AUC-ROC. Present and interpret your results. Which algorithm would you recommend, and why? (12 pts)
- c. Describe the core assumptions made by the Naïve Bayes classifier. Are these assumptions plausible in a medical feature space? What are the consequences if they are violated? (8 pts)

DATA Source: Use the Stroke Prediction Dataset (Fedesoriano, 2021) on Kaggle which contains 5,110 patient records with features including age, hypertension, heart disease, BMI, glucose level, and a binary stroke outcome with approximately 5% positive cases, making the class imbalance issue concrete and realistic. This dataset is available at: https://www.kaggle.com/datasets/fedesoriano/stroke-prediction-dataset

Problem 5. Unsupervised Learning
Dimensionality Reduction and Clustering of Survey Data

An educational survey collected responses on 40 Likert-scale items from 800 respondents. The problem requires you to identify the latent structure in the data and cluster respondents into interpretable groups.

- a. Apply Principal Component Analysis (PCA) to reduce the dimensionality of the data. How would you decide how many components to retain? Discuss the scree plot and cumulative variance explained criteria. (5 pts)
- b. Apply k-means clustering to the reduced data. Describe how you would select the optimal number of clusters k, and interpret three resulting clusters substantively in educational terms. (12 pts)
- c. Discuss one limitation of k-means clustering and explain how Non-Negative Matrix Factorization (NMF) might offer an alternative interpretation of the latent structure. (8 pts)

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DATA Source: Use the PISA 2018 Student Questionnaire Data which contains Likert-scale items on student well-being, motivation, and learning strategies across 600,000+ respondents, using a manageable subsample (e.g., U.S. respondents, ~4,800 cases) works well for in-class use. Available through the OECD at: https://www.oecd.org/pisa/data/2018database/.

Problem 6. When Should You Use Machine Learning? A Critical Analysis

A colleague proposes replacing a well-validated logistic regression model used for student early-warning identification with a Support Vector Machine (SVM), arguing that SVMs are "more advanced." Evaluate this proposal.

- a. Describe the bias-variance tradeoff. In what circumstances might the logistic regression model actually outperform the SVM despite being simpler? (5 pts)
- b. What criteria—beyond predictive accuracy—should govern the choice of model in a high-stakes educational setting? Discuss at least three considerations (e.g., interpretability, fairness, sample size). (8 pts)
- c. Your colleague also suggests using a neural network to further improve prediction. Briefly explain how neural networks differ from the "classic" supervised methods covered in this course and articulate one scenario in which a neural network would and one in which it would not be the appropriate choice. (12 pts)

Problem 6 —Use the *Open University Learning Analytics Dataset (OULAD)* (Kuzilek, Hlosta, & Zdrahal, 2017) which contains demographic, behavioral, and assessment data on 32,593 students, available at: https://analyse.kmi.open.ac.uk/open_dataset and on Kaggle. The original citation is: Kuzilek, J., Hlosta, M., & Zdrahal, Z. (2017). Open University Learning Analytics dataset. Scientific Data, 4, 170171.

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